Murali Sai Buddakkagari Venkata
AI Engineer

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SKILLS

Programming Languages: Python, SQL, JavaScript (React), C# (.NET)
AI & Machine Learning: Multi-Agent Orchestration, Agent Harness & Context Engineering, Prompt Engineering, MCP & Agent Skills, LangGraph, LangChain, Retrieval-Augmented Generation, Agent Evaluation (RAGAS), Embeddings, Vector Databases, PyTorch, MLflow
Backend & Databases: FastAPI, REST APIs, PostgreSQL, MySQL, Redis, ChromaDB, pgvector
Cloud & DevOps: Docker, Google Cloud Platform, Azure, GitHub Actions, CI/CD, Git, OpenTelemetry

EDUCATION

Stevens Institute of Technology **May 2026**
Master of Science – Applied Artificial Intelligence | GPA: 3.7 *NJ, US*

PES University **May 2024**
Bachelor of Technology – Computer Science (Specialization in ML) *KA, IND*

PROFESSIONAL EXPERIENCE

Dreamline AI – AI Engineer **Sep 2026 – Present**
Community Dreams Foundation · Remote · Data layer & admin UI (PostgreSQL, Supabase, Lovable)
- Developed a **Supabase/PostgreSQL** data layer for incentive applications using **Row Level Security** and audit triggers, and applied it to production with **0 of 269 programs altered** and all **7 acceptance criteria** met.
- Performed a **Lovable** planning pass, production baselining, and two rolled-back trial runs to catch 2 live-schema issues before a SHA-256-verified apply, and built the admin/provider requirements UI with a server-driven **"Apply Now" readiness rule**, importing **120 new programs** with 0 failures.

Querkey Inc – Data Science Engineer **Jun 2025 – May 2026**
CA, US · Multimodal AI & MLOps (Gemini, RAG, MLflow)
- Developed a **Dockerized** observability stack using **MLflow**, **FastAPI**, **ChromaDB**, and HuggingFace monitoring for real-time failure detection, reducing production ML workflow failures by **60%** across **100K+ records**.
- Developed a proof-of-concept real-time **Patient Flow Tracking System** with healthcare stakeholders using multimodal AI (**Gemini 2.5 Flash**) and **RAG**, authoring a **30-query SQL bank** to ground chatbot answers and building a **Superset REST API** integration for on-demand dashboard charts.

View Group India – Full-Stack Software Engineer **Jan 2024 – Aug 2024**
Formerly View Synergy · KA, IND · Full-stack (React, .NET, MySQL)
- Reviewed **React, .NET, and MySQL** code on a cross-functional team modernizing a live platform, root-causing 2 recurring exceptions and surfacing SQL-injection risks in a **94-issue security assessment**.
- Built and tested a **Roles/Designations CRUD feature** in MySQL, .NET, and React, and audited **Redis** usage across **6 microservices**, documenting **Entra ID** setup to support replacing custom JWT/BCrypt auth.

PROJECTS

SEC EDGAR RAG Filing Analyzer – Python, FastAPI, ChromaDB, Cloud Run
- Developed a **RAG** system on FastAPI and ChromaDB over **8,232 SEC 10-K chunks** across 5 companies, implementing **BM25 plus dense retrieval** with rank fusion, **cross-encoder reranking**, and **RBAC** information barriers, deployed on Cloud Run and verified by a **504-test CI suite** and a **54-question** evaluation (**0.853** faithfulness, **0.936** citation accuracy, **0.981** refusal correctness).
- Engineered an **agent harness** (entity detection pinning retrieval to the named filing, per-entity budgets, agentic retry, and prompt-injection and MNPI **guardrails**), exposed as an **MCP server** for Claude Desktop and Cursor.

Agent Orchestration System with Tool Use, Memory, and Human in the Loop – LangGraph
- Engineered a **multi-agent** system with **19 REST endpoints** on a 3-layer **LangGraph** hierarchy (planning supervisor, parallel specialists across dependency frontiers, reflective reviewer with a rework loop), implementing **multi-model routing** and two-tier **Redis/pgvector** memory, validated by **106/113 tests** on a live deployment.
- Built an **MCP server** with 6 agent skills, **OpenTelemetry** tracing with cost tracking, and **Human-in-the-Loop** escalation with **4 severity levels** and audit logging, following a spec-driven development process.

FilingTune: LoRA Fine-Tuning for SEC Filing QA (in progress) – PEFT, TRL, MLflow
- Building a **QLoRA fine-tuning** pipeline with PEFT, TRL, and **MLflow** to turn a small open-weight model into the citation-producing answerer for the SEC EDGAR RAG system, with a leakage-controlled data split.
- Designing the evaluation against the base model and the hosted generator on a **frozen 54-question suite**, with confidence intervals and an error taxonomy.

PUBLICATIONS AND CERTIFICATES

- Adaptive Ensemble Detection for Medical Imaging (M.S. Capstone, Stevens): developed a size-conditioned **Weighted Box Fusion** ensemble of Faster R-CNN, Mask R-CNN, and YOLO26, reaching **0.71 mAP@0.5**.
- **Best Research Paper Award**, ICICC 2024: developed WorkFit, a real-time posture monitoring system using CNN vision and flex sensors.